YEAR 11 PHYSICS, AQA GCSE, LESSON 8 OF 10

Atoms & Nuclear Radiation.

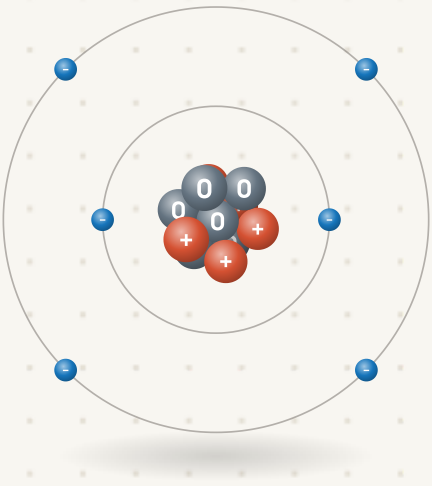
Lesson 08 Half-life and random decay

You cannot predict when one nucleus will decay. So how can we say that half of a sample of iodine-131 will be left after 8 days?

IN THIS LESSON YOU WILL

- explain why decay is random but a large sample is predictable
- define half-life in two ways
- find the half-life from a decay curve
- plot a decay curve from data

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

How this booklet works.

This is the booklet for **Lesson 8** of the Atoms and Nuclear Radiation unit. It builds on the lessons before it; the "Do now" questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
Explanations, diagrams and worked examples. Read this again when you revise.

TRY IT
Practice that builds up: first with support, then on your own, then a challenge.

LAB
Record what you find in a simulation or a model experiment.

EXAM CORNER
AQA-style questions with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an "I can" check to colour in.

! **Revising at home?** (1) Read the Learn it pages. (2) Cover them and answer the "Cover the page" questions in Revise it. (3) Check with the quick answers at the back. (4) Go through the matching slides in the teaching deck, and try the simulations.

The route through the unit (this lesson is highlighted)

01 Inside the atom

02 Isotopes and ions

03 From plum pudding to the nucleus

04 Energy levels and the nucleus

05 Radioactive decay

06 Nuclear equations

07 Properties and uses of radiation

08 Half-life and random decay
Random but predictable, defining half-life, reading a decay curve (slides 35–37)

09 Half-life calculations

10 Contamination and irradiation

Teaching deck: **panphy.app/gcsephy/decks/atoms-and-radiation**. Simulations: **panphy.app/simulations/atomic_models.html** and **panphy.app/simulations/nuclear_decay.html**

LESSON

08

Half-life and random decay.

AQA Physics 4.4.2.3; Combined Science 6.4.2.3

BIG QUESTION

You cannot predict when one nucleus will decay. So how can we say that half of a sample of iodine-131 will be left after 8 days?

BY THE END YOU CAN

- explain why decay is random but a large sample is predictable
- define half-life in two ways
- find the half-life from a decay curve
- plot a decay curve from data

half-life | random | decay curve | sample | count rate | activity

SLIDES AND SIMULATIONS

Teaching deck, slides 35–37

Half-life, using half-life and random decay.

panphy.app/gcsephy/decks/atoms-and-radiation/#35

Nuclear Decay simulation

panphy.app/simulations/nuclear_decay.html

RETRIEVE

Do now

From earlier lessons. No notes!

1 Which radiation is stopped by paper?

Alpha.

2 What does a count rate measure?

The number of decays a detector records each second.

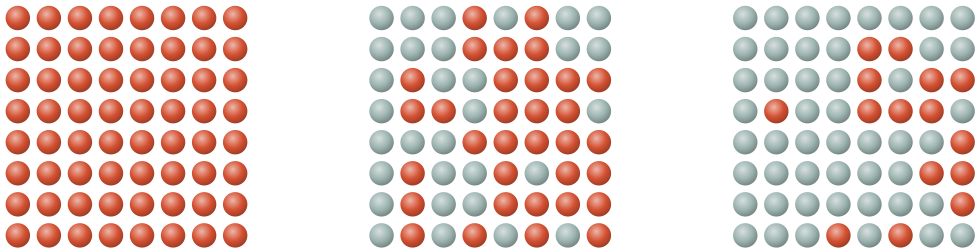
3 What is the unit of activity?

Becquerel (Bq).

EXPLAIN

Random, but predictable

Random, but predictable



start
64 undecayed

after 1 half-life
33 undecayed (≈ 32)

after 2 half-lives
15 undecayed (≈ 16)

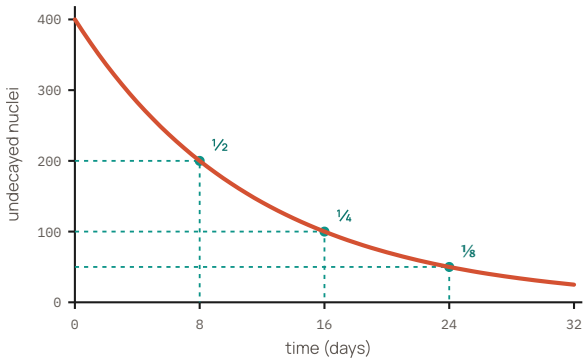
Every unstable nucleus of an isotope has the **same chance** of decaying in a given time. You cannot say which one will decay next, or when. But in a **large sample**, the number that decay in a given time is very predictable. In a small sample, the numbers vary more.

THINK OF COINS

Throw 100 coins and remove every head. About half are left. Throw the rest and remove the heads again: about a quarter are left. You cannot predict any one coin, but you can predict the group.

Where the model breaks: coins only 'decay' when you throw them, but nuclei decay at any moment; and real samples have billions of nuclei.

Half-life



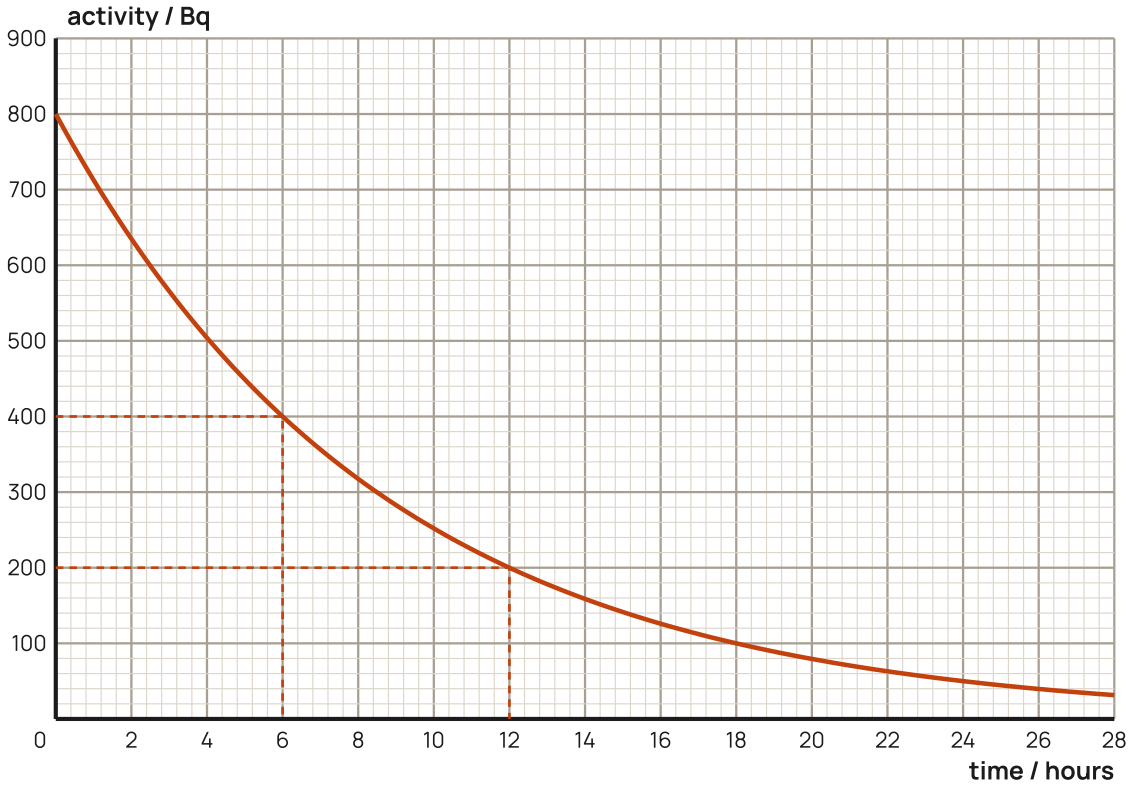
The **half-life** of a radioactive isotope is:

- the time it takes for the **number of unstable nuclei** in a sample to **halve**, or
- the time it takes for the **count rate** (or activity) from a sample to fall to **half** its initial level.

Each halving takes the same time. Iodine-131 has a half-life of about 8 days.

Finding the half-life from a graph

1. Read the **starting value**: here 800 Bq.
2. **Halve it**: 400 Bq.
3. Read **across** from 400 Bq to the curve, then **down** to the time axis: 6 hours.
4. **Check** with a second halving: 400 → 200 Bq also takes 6 hours.

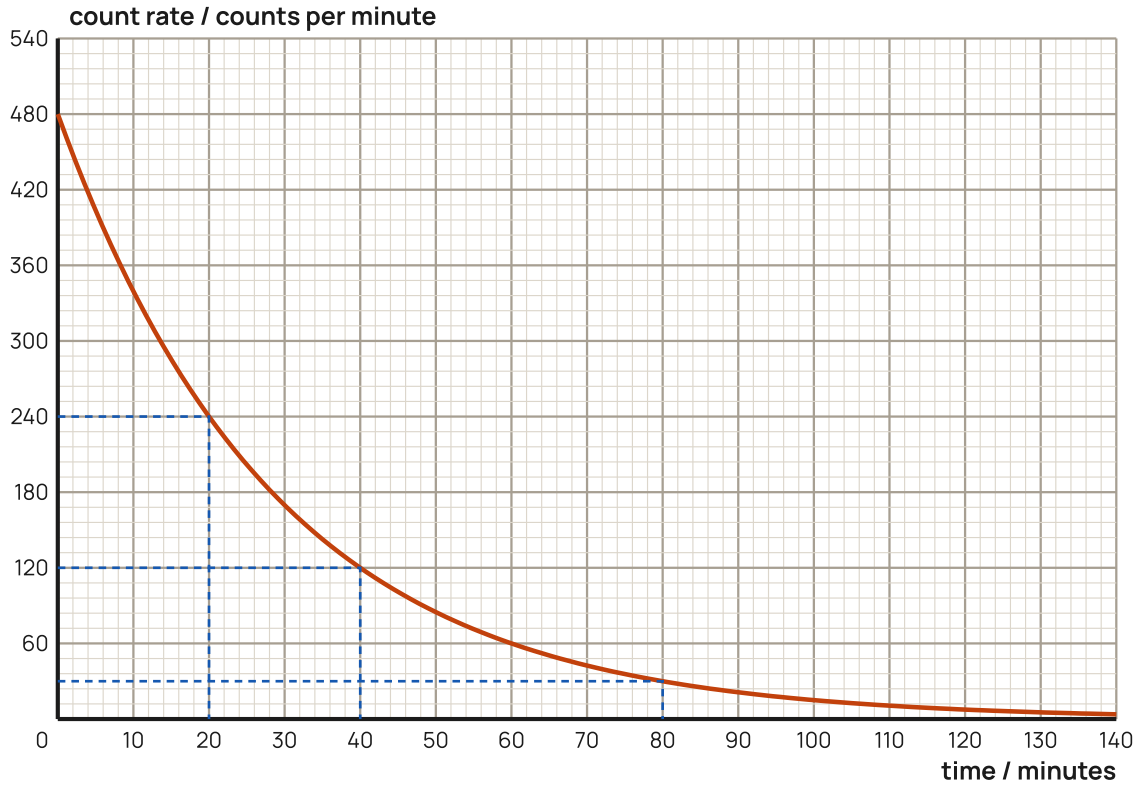


! Half-life = **6 hours**. Every halving takes the same time: 800 → 400 → 200 Bq at 6 and 12 hours.

PRACTISE

Try it

1 The graph shows how the count rate from a sample changes. Use it to find:



(A) HALF-LIFE	480 → 240 at 20 minutes: half-life = 20 min
(B) COUNT RATE AT 40 MIN	120 counts per minute
(C) TIME TO REACH 30	80 minutes (4 half-lives)

2 The half-life of a sample is 5 days. Which statement is correct? Tick **one** box.

- ☐ All the nuclei will have decayed after 10 days.
- ■ About half of the unstable nuclei will decay in the next 5 days.
- ☐ Each nucleus lasts exactly 5 days before it decays.
- ☐ After 5 days the sample is no longer radioactive.

3 SUPPORTED A sample has 1600 unstable nuclei. Its half-life is 3 hours. Complete the table.

Time / hours	0	3	6	9	12
Unstable nuclei left	1600	800	400	200	100

4 Explain what is wrong with Leo's statement.



After one half-life half is left; after two half-lives, a quarter (half of a half) is left, not zero.

5 A student has a sample of only 100 nuclei. After one half-life, 44 are left, not 50. Explain why.

Decay is random. In a small sample the number that decay varies a lot around the expected half; a larger sample would be closer to 50%.

INVESTIGATE

Virtual lab: modelling half-life

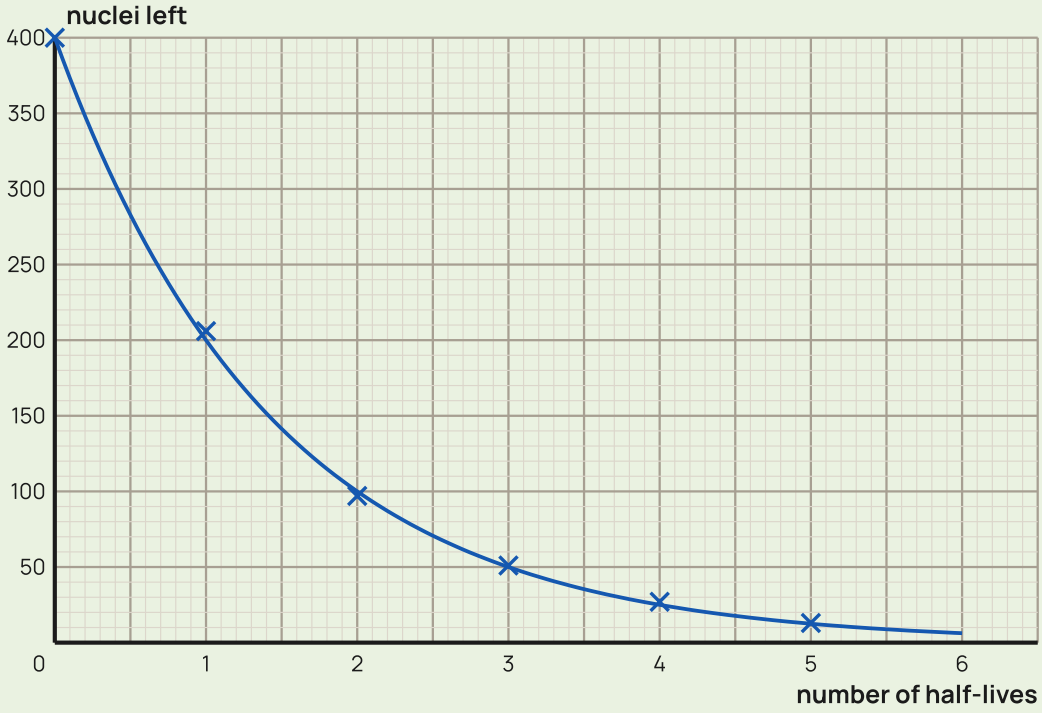
Nuclear Decay simulation, tab 04 (Half-life). No computer? Use 100 coins: each throw is one half-life; remove the heads.

- Open tab **04 Half-life**. Choose **Iodine-131** and **400 nuclei**.
- Before each step, write the number you expect. Then press **Run one half-life** and record the number left.
- Repeat until 5 half-lives have passed.

6 Record your results.

Half-lives	0	1	2	3	4	5
Expected number left	400	200	100	50	25	12.5
Number left (your run)	400	e.g. 206	e.g. 97	e.g. 51	e.g. 27	e.g. 13

7 Plot your results. Draw a smooth curve of best fit.



8 Now run the simulation again with **100 nuclei**. How is the graph different? Explain why.

It is more ragged: the numbers left are further from exact halves. With fewer nuclei, the random nature of decay has a bigger effect.

AQA-STYLE

Exam corner

Write in full sentences. Show every step of a calculation.

Get the marks: One definition is enough for 1 mark, but it must say 'half' and what is halving.

9 Define the half-life of a radioactive isotope. **[1 mark]**

The time taken for the number of unstable nuclei in a sample (or the count rate) to halve.

10 Explain why the activity of a radioactive source decreases over time. **[2 marks]**

As nuclei decay, there are fewer unstable nuclei left (1), so there are fewer decays each second (1).

REVIEW LATER

Revise it

Use this box after the lesson and before tests.

SUMMARY

- Decay is random: we cannot predict which nucleus decays, or when.
- A large sample is predictable; small samples vary more.
- Half-life: the time for the number of unstable nuclei, or the count rate, to halve.
- From a graph: start value → halve it → across to the curve → down to the time.

COVER THE PAGE AND ANSWER

1 Define half-life.

2 What fraction is left after 2 half-lives?

3 Why are small samples less predictable?

4 What is the half-life of iodine-131?

I CAN…

- explain why decay is random but predictable
- define half-life in two ways
- find a half-life from a decay curve
- plot a decay curve from data

REFERENCE

Toolkit.

The facts and methods you have met so far. Learn these: exam questions expect you to recall them without a data sheet.

The three particles

Lesson 1

Particle	Charge	Mass	Where
Proton	+1	1	nucleus
Neutron	0	1	nucleus
Electron	−1	very small	energy levels

Reading the symbol

Lesson 1

$^{A}_{Z}X$ e.g. $^{23}_{11}Na$

- Z, atomic number = protons
- A, mass number = protons + neutrons
- neutrons = A − Z; in an atom, electrons = Z
- Atom radius ≈ 1×10^{-10} m; nucleus radius < 1/10 000 of that

Isotopes and ions

Lesson 2

- **Isotopes**: same number of protons, different numbers of neutrons.
- **Positive ion**: an atom that has lost one or more outer electrons. The nucleus is unchanged.

Alpha scattering

Lesson 3

Observation	Conclusion
most straight through	mostly empty space
some deflected	positive centre
a very few bounced back	mass in a tiny centre

Models and energy levels

Lesson 4

- 1897 electron → 1904 plum pudding → 1911 nuclear → 1913 Bohr → 1932 neutron (Chadwick)
- **Absorb** EM radiation → higher level, further out.
- **Emit** EM radiation → lower level, closer in.

Nuclear radiation

Lesson 5

- Unstable nuclei decay at **random** to become more stable.
- **Activity**: decays per second, in becquerel (Bq). **Count rate**: decays recorded per second by a detector.
- α: 2p + 2n; β: fast electron, a neutron → proton; γ: EM radiation; n: a neutron

Nuclear equations

Lesson 6

Emitted	Symbol	Mass no.	Atomic no.
alpha	$^{4}_{2}\mathrm{He}$	− 4	− 2
beta	$^{0}_{-1}\mathrm{e}$	no change	+ 1
gamma	γ	no change	no change
neutron	$^{1}_{0}\mathrm{n}$	− 1	no change

Top numbers balance; bottom numbers balance.

$$^{226}_{88}\mathrm{Ra} \rightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He}$$

$$^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\mathrm{e}$$

Properties of radiation

Lesson 7

	Alpha	Beta	Gamma
Ionising	strongly	moderately	weakly
Range in air	a few cm	about 1 m	very far
Stopped by	paper, skin	a few mm of aluminium	reduced by thick lead or concrete
Typical use	smoke alarm	thickness gauge	medical tracer

More ionising → less penetrating.

Half-life from a graph

Lesson 8

1. Read the start value.
2. Halve it.
3. Across to the curve, down to the time axis.
4. Check with a second halving.

REFERENCE

Key words from this lesson.

Learn the meaning and spelling. In exams, precise words earn marks: say “nucleus”, not “middle”, and “unstable nucleus”, not “unstable atom”.

Half-life: The time for the number of unstable nuclei in a sample, or the count rate, to halve.

Random: Impossible to predict for one nucleus: which one decays next, or when.

CHECK YOURSELF

Quick answers.

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

DO NOW

1. Alpha.
2. The number of decays a detector records each second.
3. Becquerel (Bq).

COVER THE PAGE

1. The time for the number of unstable nuclei (or the count rate) to halve.
2. A quarter (1/4).
3. Decay is random, so the numbers vary more around the expected value.
4. About 8 days.